Carla Becker

Program & Technical Leadership

- **Program & Execution**
  Technical program management, roadmap definition, early-stage project ramp-up; Cross-functional coordination across ML, software, hardware, and operations; Vendor management, SOWs, budgeting, and delivery against milestones.
- **AI/ML & Systems**
  Machine learning model evaluation and lifecycle management; experimentation pipelines, data quality, and deployment tradeoffs under compute and latency constraints; Simulation-ML hybrid systems (physics-based + learned models).
- **Technical Stack**
  Python, basic SQL and C++, Flask, Snowflake, Docker/Kubernetes, Unix/Bash; Data visualization, internal tooling, automation, and web apps.

Contact

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Education

University of California Berkeley
Ph.D., Mechanical Engineering;
Expected: May 2026
M.S., Mechanical Engineering, 2024
M.S., Materials Science & Engineering, 2022

Harvey Mudd College
B.S. Physics, B.S. Chemistry, 2018

Professional Experience

- **Technical Lead, Design for Manufacturing**, Apple Inc. — *February 2025 - August 2025*
  - Owned end-to-end development of an internal AI-enabled platform for component-level data ingestion, visualization, and live editing, used across DFM workflows.
  - Coordinated ML modeling, backend infrastructure (Snowflake, Flask), deployment (Docker/K8s), and user feedback loops.
  - Led evaluation of 18 ML approaches for solder joint crack severity prediction, framing tradeoffs between accuracy, interpretability, and compute cost. Supported rapid iteration and deployment.
  - Proposed a novel hybrid FEA + neural-network approach to reduce system-level simulation time by orders of magnitude and road-mapped its development.
- **Program Manager, Product Engineer**, Sakuu Corp. — *May 2018 - June 2022*
  - Led a $300K, 1.5-year internal program to design and deploy a browser-based ML and analytics platform for battery development.
  - Defined scope, success metrics, and 3-year roadmap for a new modeling & simulation business unit.
  - Owned vendor selection, SOWs, and stakeholder alignment across engineering, operations, and leadership.
  - Managed a 10-person technical team and multiple interns, balancing delivery timelines with R&D uncertainty.
- **Business Operations Management**, Intel Corporation — *June 2022 - May 2023*
  - Streamlined hiring and finance workflows for the large Developer Software Engineering org by automating headcount reporting, OKR tracking, and requisition generation.
  - Reduced manual reporting overhead and improved leadership visibility using Python, PowerBI, and automation pipelines.
- **Program Manager & Technical Lead**, UC Berkeley — *June 2022 - Present*
  - **Program Manager, DEWA** Owned execution of an international engineering certificate program administered jointly by UC Berkeley and Dubai Electricity & Water Authority (DEWA).
  - Coordinated faculty, registrars, TAs, and DEWA leadership across time zones, managing schedules, enrollment, grading workflows, and escalations.
  - **Technical Lead, Capstone Connect** Led design and delivery of an automated matching platform supporting hundreds of industry-sponsored MEng capstone projects.
  - Translated multi-stakeholder requirements into ranking logic and system constraints, replacing manual workflows.

Research Experience

- **Graduate Student Researcher**, UC Berkeley — *May 2021 - Present*
  - Led development of ML-driven optimization and simulation tools across materials discovery ((The Materials Project) and agricultural systems (AI Institute for Next-Gen Food Systems), spanning ODE models, ray tracing, and scalable experimentation pipelines.
  - Experience translating research prototypes into usable internal tools and platforms (APIs, AWS, testing, performance optimization in Python).
  - Outstanding Graduate Student Instructor 2024, Modeling and Simulation of Advanced Manufacturing Systems.